Fischgrund et. al (2018) conducted a randomized double blind sham controlled multi-center study to evaluate the safety and efficacy of radiofrequency (RF) ablation of the basivertebral nerve (BVN) for the treatment of chronic low back pain (CLBP) in a Food and Drug Administration approved Investigational Device Exemption trial. A total of 225 patients diagnosed with CLBP were randomized to either a sham (78 patients) or treatment (147 patients) intervention. The mean age within the study was 47 years (range 25-69) and the mean baseline ODI was 42. All patients had Type I or Type II Modic changes of the treated vertebral bodies. Patients were evaluated preoperatively, and at 2 weeks, 6 weeks and 3, 6 and 12 months postoperatively. The primary endpoint was the comparative change in ODI from baseline to 3 months. At 3 months, the average ODI in the treatment arm decreased 20.5 points, as compared to a 15.2 point decrease in the sham arm ($p = 0.019$, per-protocol population). A responder analysis based on ODI decrease ≥ 10 points showed that 75.6% of patients in the treatment arm as compared to 55.3% in the sham control arm exhibited a clinically meaningful improvement at 3 months. Conclusion: Patients treated with RF ablation of the BVN for CLBP exhibited significantly greater improvement in ODI at 3 months and a higher responder rate than sham treated controls. BVN ablation represents a potential minimally invasive treatment for the relief of chronic low back pain.

In a follow up study by Fischgurd et. al (2020), the long-term outcomes following intraosseous basivertebral nerve ablation for the treatment of chronic low back pain were assessed: 5-year treatment arm results from a prospective randomized double-blind sham-controlled multi-center study. In total, 117 US patients were treated successfully with BVN ablation. Patient-reported outcomes of post ablation treatments, and patient satisfaction were collected at a minimum of 5-years following BVN ablation. Comparisons between the post ablation and baseline values were made using an analysis of covariance with alpha 0.05.

Results: Of the 117 US treated patients 100 (85%) were available for review with a mean follow-up of 6.4 years (5.4-7.8 years). Mean Oswestry Disability Index (ODI) score improved from 42.81 to 16.86 at 5-year follow-up, a reduction of 25.95 points ($p < 0.001$). Mean reduction in Visual Analogue Scale (VAS) pain score was 4.38 points (baseline of 6.74, $p < 0.001$). In total, 66% of patients reported a > 50% reduction in pain, 47% reported a > 75% reduction in pain, and 34% of patients reported complete pain resolution. Composite responder rate using thresholds of ≥ 15-point ODI and ≥ 2-point VAS for function and pain at 5 years was 75%.

Conclusion: CLBP patients treated with BVN ablation exhibit sustained clinical improvements in function and pain with high responder rates at a mean of 6.4 years following treatment. BVN ablation is a durable, minimally invasive treatment for vertebrogenic CLBP.

In a prospective, randomized, multicenter study of intraosseous basivertebral nerve ablation for the treatment of chronic low back pain conducted by Khalil et. al (2019), comparison of the effectiveness of intraosseous RF ablation of the BVN to standard care for the treatment of chronic LBP in a specific subgroup of patients suspected to have vertebrogenic related symptomatology was evaluated. A total of